Quantifying Uncertainty

Outline

- Uncertainty
- Probability
- Syntax and Semantics
- Inference
- Independence and Bayes' Rule

Why?

- **Real-World Challenges:** In many AI applications, information is incomplete, noisy, or uncertain.
- **Goal:** AI systems need to reason and make decisions despite this uncertainty.
- **Solution:** Probability theory and Bayesian reasoning are key tools used to model uncertainty.
- **Applications:** Self-driving cars, medical diagnosis systems, and weather forecasting.

Probability Notation

Syntax

- In probability, a real-valued function, defined over the sample space of a random experiment, is called a random variable. That is, **the values of the random variable correspond to the outcomes of the random experiment**. Random variables could be either discrete or continuous.

  Boolean random variables

  e.g., *Cavity* (do I have a cavity?): True or False | 0 or 1

  Discrete random variables

  e.g., *Weather* is one of <*sunny,rainy,cloudy,snow*>

  Domain values must be exhaustive and mutually exclusive

- Elementary proposition constructed by assignment of a value to a random variable: e.g., *Weather = sunny*, *Cavity = false*
- Complex propositions formed from elementary propositions and standard logical connectives e.g., *Weather = sunny* $\vee$ *Cavity = false*

Syntax

- **Atomic event:** A **complete** specification of the state of the world about which the agent is uncertain
-

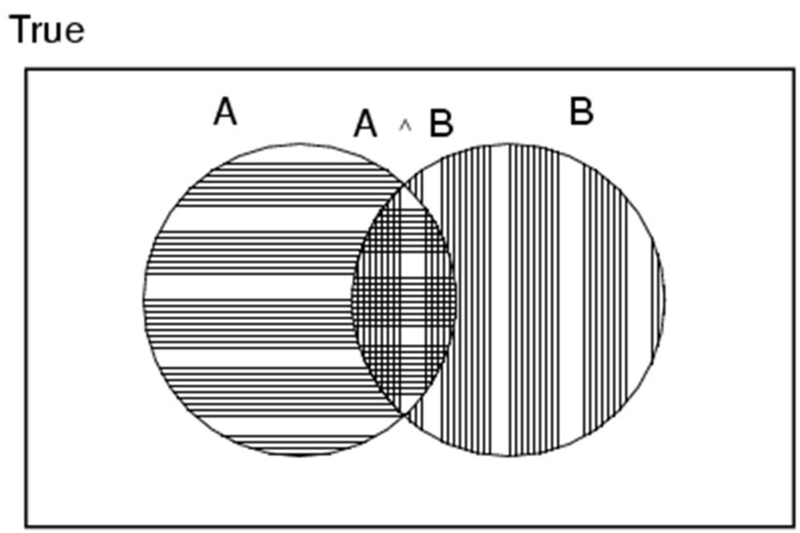
  E.g., if the world consists of only two Boolean variables *Cavity* and *Toothache*, then there are 4 distinct atomic events:

  $Cavity = false \wedge Toothache = false$
  $Cavity = false \wedge Toothache = true$
  $Cavity = true \wedge Toothache = false$
  $Cavity = true \wedge Toothache = true$

- Atomic events are mutually exclusive and exhaustive

Axioms of probability

- For any propositions *A, B*
- $0 \leq P(A) \leq 1$
  - P(*true*) = 1 and P(*false*) = 0
  - $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$



Prior probability

- **Prior or unconditional probabilities of propositions**
- e.g., P(*Cavity* = true) = 0.1 and P(*Weather* = sunny) = 0.72 correspond to belief prior to arrival of any (new) evidence

- **Probability distribution** gives values for all possible assignments:
- **P**(*Weather*) = <0.72, 0.1, 0.08, 0.1> (normalized, i.e., sums to 1)

- **Joint probability distribution** for a set of random variables gives the probability of every atomic event on those random variables
- **P**(*Weather,Cavity*) = a 4 × 2 matrix of values:

Cavity / *Weather*	sunny	rainy	cloudy	snow
Cavity = true	0.144	0.02	0.016	0.02
Cavity = false	0.576	0.08	0.064	0.08

- **Every question about a domain can be answered by the joint distribution**

Conditional probability

- **Conditional or posterior probabilities**
-

  e.g., P(*cavity* | *toothache*) = 0.8

  i.e., given that *toothache* is all I know

- If we know more, e.g., *cavity* is also given, then we have
-

  P(*cavity* | *toothache,cavity*) = 1

- New evidence may be irrelevant, allowing simplification, e.g.,
-

  P(*cavity* | *toothache, sunny*) = P(*cavity* | *toothache*) = 0.8
- This kind of inference, sanctioned by domain knowledge, is crucial
-

Conditional probability

- Definition of conditional probability:
- $$P(a \mid b) = \frac{P(a \wedge b)}{P(b)} \qquad \textit{where} \qquad P(b) > 0$$

- **Product rule** gives an alternative formulation:
- $$P(a \wedge b) = P(a \mid b)\, P(b) = P(b \mid a)\, P(a)$$

P(A): Prior probability is the initial probability of an event or hypothesis A before considering any new evidence or information.

P(A|B): Posterior probability is the updated probability of an event or hypothesis A after taking into account new evidence B.

Axioms of Probability

- Andrey Kolmogorov's three axioms

1. All probabilities are between 0 and 1

$$0 \leqq P(a) \leqq 1$$

2. Necessarily true is 1, necessarily false is 0

$$P(true)=1, \qquad P(false)=0$$

3. Disjunction

$$P(a \vee b)=P(a) + P(b) - P(a \wedge b)$$

Inference Using Full Joint Distributions

A **Full Joint Distribution** is a table or formula that gives the probability of every possible combination of the values of random variables. In probability theory and statistics, it defines the probabilities for all possible outcomes (jointly) of a set of random variables.

Inference by enumeration

- Start with the joint probability distribution:
-

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

Inference by enumeration

- Start with the joint probability distribution:
-

	toothache		¬ *toothache*	
	catch	¬ *catch*	*catch*	¬ *catch*
cavity	.108	.012	.072	.008
¬ *cavity*	.016	.064	.144	.576

- P(*toothache*) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2
-

Inference by enumeration

- Start with the joint probability distribution:
-

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	**.108**	**.012**	**.072**	**.008**
¬ cavity	**.016**	**.064**	**.144**	**.576**

- P(cavity $\vee$*toothache*) = 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 = 0.28
-

Inference by enumeration

- Start with the joint probability distribution:
-

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

- Can also compute conditional probabilities:
-

$$P(\neg cavity \mid toothache) = \frac{P(\neg cavity \wedge toothache)}{P(toothache)}$$

$$= \frac{0.016+0.064}{0.108 + 0.012 + 0.016 + 0.064}$$

$$= 0.4$$

Complexity of Enumeration

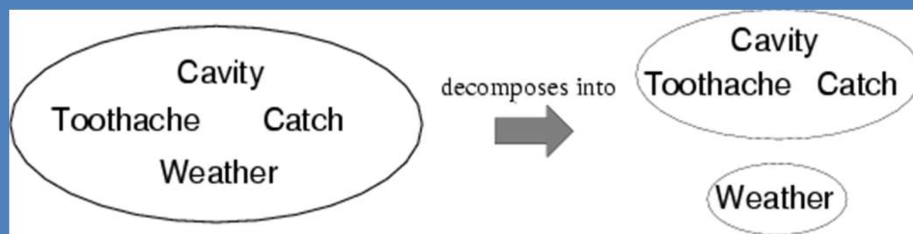
- Worst case time: $O(d^n)$
  - Where d = max arity
  - And n = number of random variables
- Space complexity also $O(d^n)$
  - Size of joint distribution

Summary: In probability theory and statistics, when dealing with multiple random variables, we often need to consider their joint probability distribution. However, specifying the full joint distribution can be computationally expensive and require a large amount of data. **Independence** and **conditional independence relationships can significantly simplify this process**.

Independence

Independence

- *A* and *B* are independent iff
  $\mathbf{P}(A|B) = \mathbf{P}(A)$ or $\mathbf{P}(B|A) = \mathbf{P}(B)$ or $\mathbf{P}(A, B) = \mathbf{P}(A)\,\mathbf{P}(B)$



$\mathbf{P}(\mathit{Toothache, Catch, Cavity, Weather})$
$= \mathbf{P}(\mathit{Toothache, Catch, Cavity})\,\mathbf{P}(\mathit{Weather})$

- 32 entries reduced to 12; for *n* independent biased coins, $O(2^n) \rightarrow O(n)$
- Absolute independence powerful but rare
- Dentistry is a large field with hundreds of variables, none of which are independent. What to do?

The **chain rule** in probability allows us to break down the joint probability distribution into a product of conditional probabilities. When applied to a **full joint distribution**, it expresses the probability of a set of random variables $X_1, X_2, \ldots, X_n$ as a product of conditional probabilities, starting with the first variable and conditioning each successive variable on the previous ones.

Chain Rule Formula:

For a set of random variables $X_1, X_2, \ldots, X_n$, the full joint distribution can be expressed as:

$$P(X_1, X_2, \ldots, X_n) = P(X_1) \cdot P(X_2|X_1) \cdot P(X_3|X_1, X_2) \cdot \cdots \cdot P(X_n|X_1, X_2, \ldots, X_{n-1})$$

In other words, each term is the conditional probability of one variable, given all the previous variables.

- **In most cases, the use of conditional independence reduces the size of the representation of the joint distribution from exponential in *n* to linear in *n*.**

Example:

Let's say we have three random variables X_1, X_2, X_3, the full joint distribution $P(X_1, X_2, X_3)$ can be written using the chain rule as:

$$P(X_1, X_2, X_3) = P(X_1) \cdot P(X_2|X_1) \cdot P(X_3|X_1, X_2)$$

This expresses the joint distribution as a series of conditional probabilities.

Why Use the Chain Rule?

The chain rule is especially useful in cases where the joint distribution is difficult to compute directly, but the conditional probabilities are easier to compute. This is often the case in Bayesian networks and other probabilistic models where dependencies between variables are structured.

Bayes' Theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- $P(A|B)$ is the **posterior probability**: the probability of class A given that feature B occurred.
- $P(B|A)$ is the **likelihood**: the probability of feature B given that class A occurred.
- $P(A)$ is the **prior probability**: the overall probability of class A (without knowing feature B).
- $P(B)$ is the **evidence**: the overall probability of feature B occurring.

Bayes' Rule

- Product rule $P(a \wedge b) = P(a \mid b)\, P(b) = P(b \mid a)\, P(a)$
- $\Rightarrow$ **Bayes' rule:** $$P(a \mid b) = \frac{P(b \mid a)\, P(a)}{P(b)}$$

- or in distribution form
- $$\mathbf{P}(Y|X) = \mathbf{P}(X|Y)\, \mathbf{P}(Y) / \mathbf{P}(X) = \alpha \mathbf{P}(X|Y)\, \mathbf{P}(Y)$$

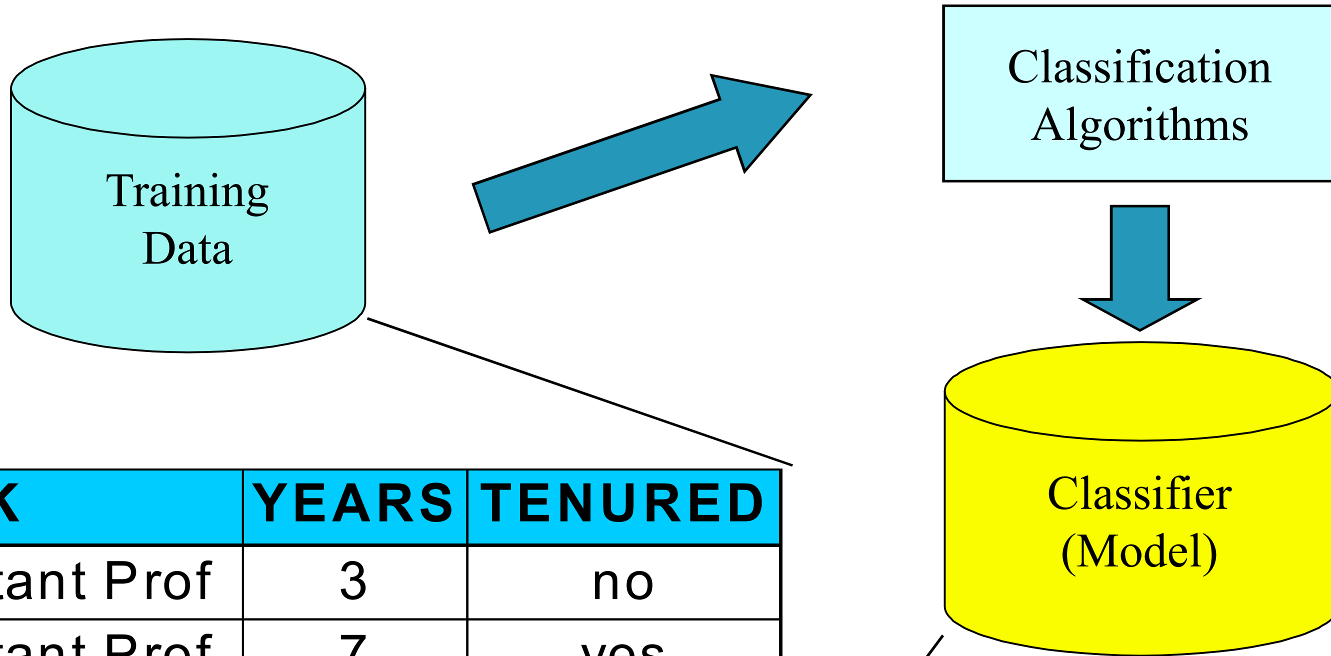
- Useful for assessing diagnostic probability from causal probability:
- **P(Cause|Effect) = P(Effect|Cause) P(Cause) / P(Effect)**

  - E.g., let *M* be meningitis, *S* be stiff neck:
  - $$P(m|s) = P(s|m)\, P(m) / P(s) = 0.5 \times 0.00002 / 0.05 = 0.0002$$

Summary

- Probability is a rigorous formalism for uncertain knowledge
- **Joint probability distribution** specifies probability of every **atomic event**
- Queries can be answered by summing over atomic events
- For nontrivial domains, we must find a way to reduce the joint size
- **Independence** and **conditional independence** provide the tools

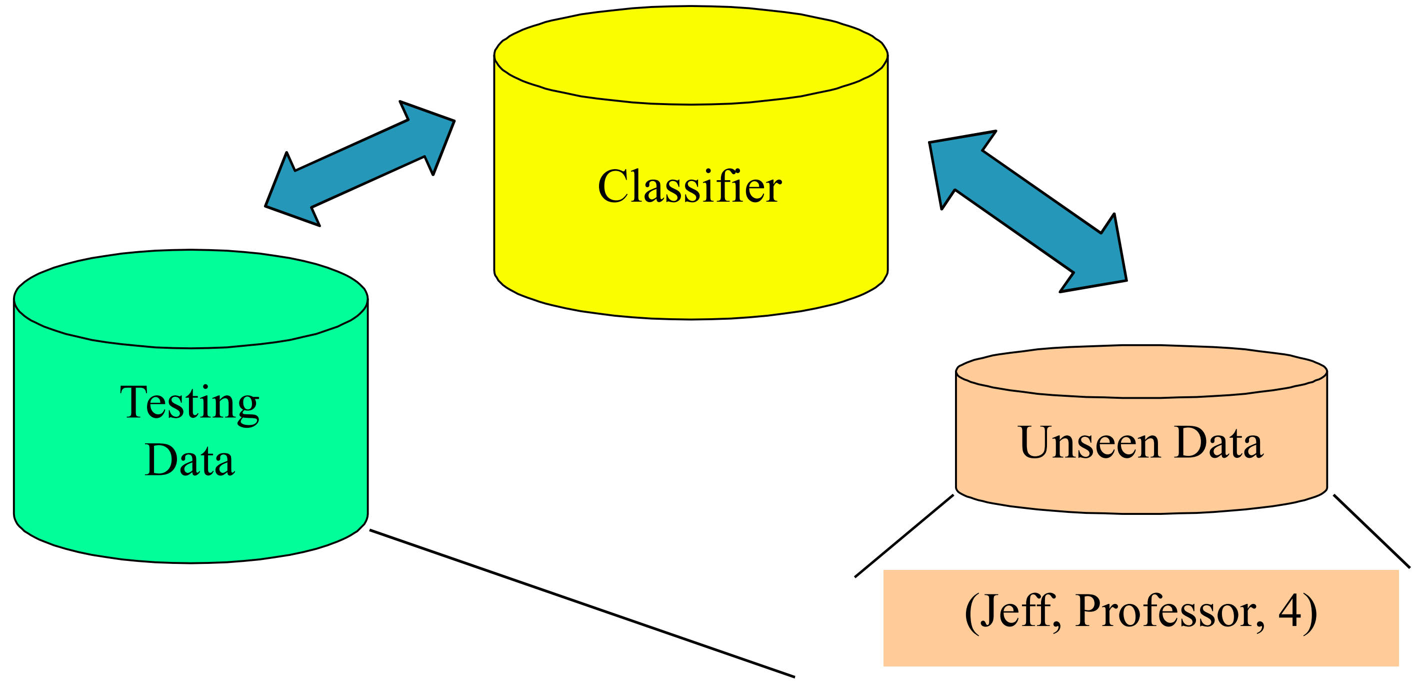
Classification Process: Model Construction



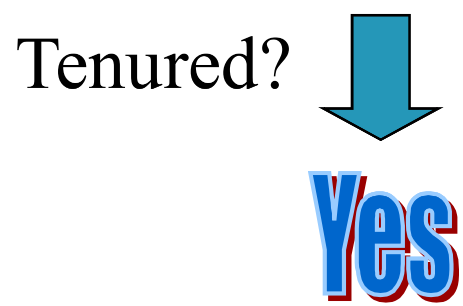
NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

IF rank = 'professor'
OR years > 6
THEN tenured = 'yes'

Classification Process : Use the Model in Prediction



NAME	RANK	YEARS	TENURED
Tom	Assistant Prof	2	no
Merlisa	Associate Prof	7	no
George	Professor	5	yes
Joseph	Assistant Prof	7	yes



Bayesian classification

- The classification problem may be formalized using a-posteriori probabilities:
- $P(C|X)$ = prob. that the sample tuple $X = \langle x_1,\ldots,x_k \rangle$ is of class C.
- E.g. P(class=N | outlook=sunny,windy=true,…)
- Idea: assign to sample X the class label C such that $P(C|X)$ is maximal

Estimating a-posteriori probabilities

- Bayes theorem:

$$P(C|X) = P(X|C) \cdot P(C) \,/\, P(X)$$

- $P(X)$ is constant for all classes
- $P(C)$ = relative freq of class C samples
- C such that $P(C|X)$ is maximum =

  C such that $P(X|C) \cdot P(C)$ is maximum
- Problem: computing $P(X|C)$ is unfeasible!

Naïve Bayesian Classification

- Naïve assumption: attribute independence

  $$P(x_1,\ldots,x_k|C) = P(x_1|C)\cdot\ldots\cdot P(x_k|C)$$

- If i-th attribute is categorical:
  $P(x_i|C)$ is estimated as the relative freq of samples having value x_i as i-th attribute in class C
- If i-th attribute is continuous:
  $P(x_i|C)$ is estimated thru a Gaussian density function
- Computationally easy in both cases

Play-tennis example: estimating $P(x_i|C)$

Outlook	Temperature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
rain	cool	normal	true	N
overcast	cool	normal	true	P
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
overcast	hot	normal	false	P
rain	mild	high	true	N

outlook	
P(sunny|p) = 2/9	**P(sunny|n) = 3/5**
P(overcast|p) = 4/9	**P(overcast|n) = 0**
P(rain|p) = 3/9	**P(rain|n) = 2/5**
temperature	
P(hot|p) = 2/9	**P(hot|n) = 2/5**
P(mild|p) = 4/9	**P(mild|n) = 2/5**
P(cool|p) = 3/9	**P(cool|n) = 1/5**
humidity	
P(high|p) = 3/9	**P(high|n) = 4/5**
P(normal|p) = 6/9	**P(normal|n) = 2/5**
windy	
P(true|p) = 3/9	**P(true|n) = 3/5**
P(false|p) = 6/9	**P(false|n) = 2/5**

P(p) = 9/14
P(n) = 5/14

Play-tennis example: classifying X

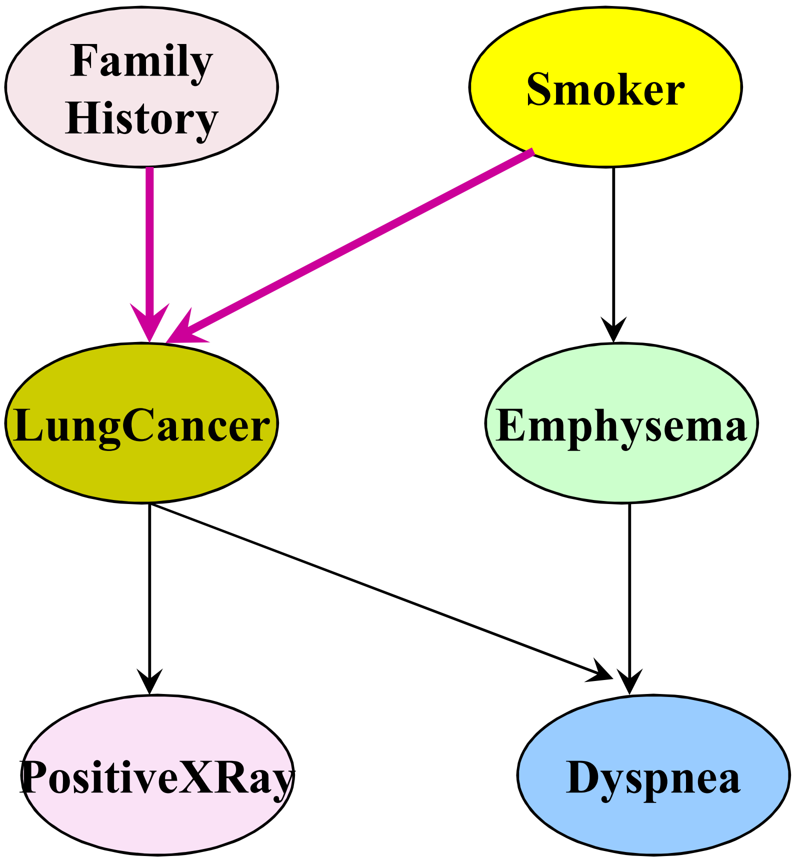
- An unseen sample X = <rain, hot, high, false>

- $P(X|p) \cdot P(p) =$
  $P(rain|p) \cdot P(hot|p) \cdot P(high|p) \cdot P(false|p) \cdot P(p) =$
  $3/9 \cdot 2/9 \cdot 3/9 \cdot 6/9 \cdot 9/14 = 0.010582$
- $P(X|n) \cdot P(n) =$
  $P(rain|n) \cdot P(hot|n) \cdot P(high|n) \cdot P(false|n) \cdot P(n) =$
  $2/5 \cdot 2/5 \cdot 4/5 \cdot 2/5 \cdot 5/14 = 0.018286$
- Sample X is classified in class n (don't play)

The independence hypothesis…

- … makes computation possible
- … yields optimal classifiers when satisfied
- … but is seldom satisfied in practice, as attributes (variables) are often correlated.
- Attempts to overcome this limitation:
  - Bayesian networks, that combine Bayesian reasoning with causal relationships between attributes
  - Decision trees, that reason on one attribute at the time, considering most important attributes first

Bayesian Belief Networks (I)



Bayesian Belief Networks

	(FH, S)	(FH, ~S)	(~FH, S)	(~FH, ~S)
LC	0.8	0.5	0.7	0.1
~LC	0.2	0.5	0.3	0.9

The conditional probability table for the variable LungCancer

Bayesian Belief Networks (II)

- Bayesian belief network allows a *subset* of the variables conditionally independent
- A graphical model of causal relationships
- Several cases of learning Bayesian belief networks
  - Given both network structure and all the variables: easy
  - Given network structure but only some variables
  - When the network structure is not known in advance